POKHARA UNIVERSITY

Tutorial sheets

on

Engineering Drawing (MEC 109.2)

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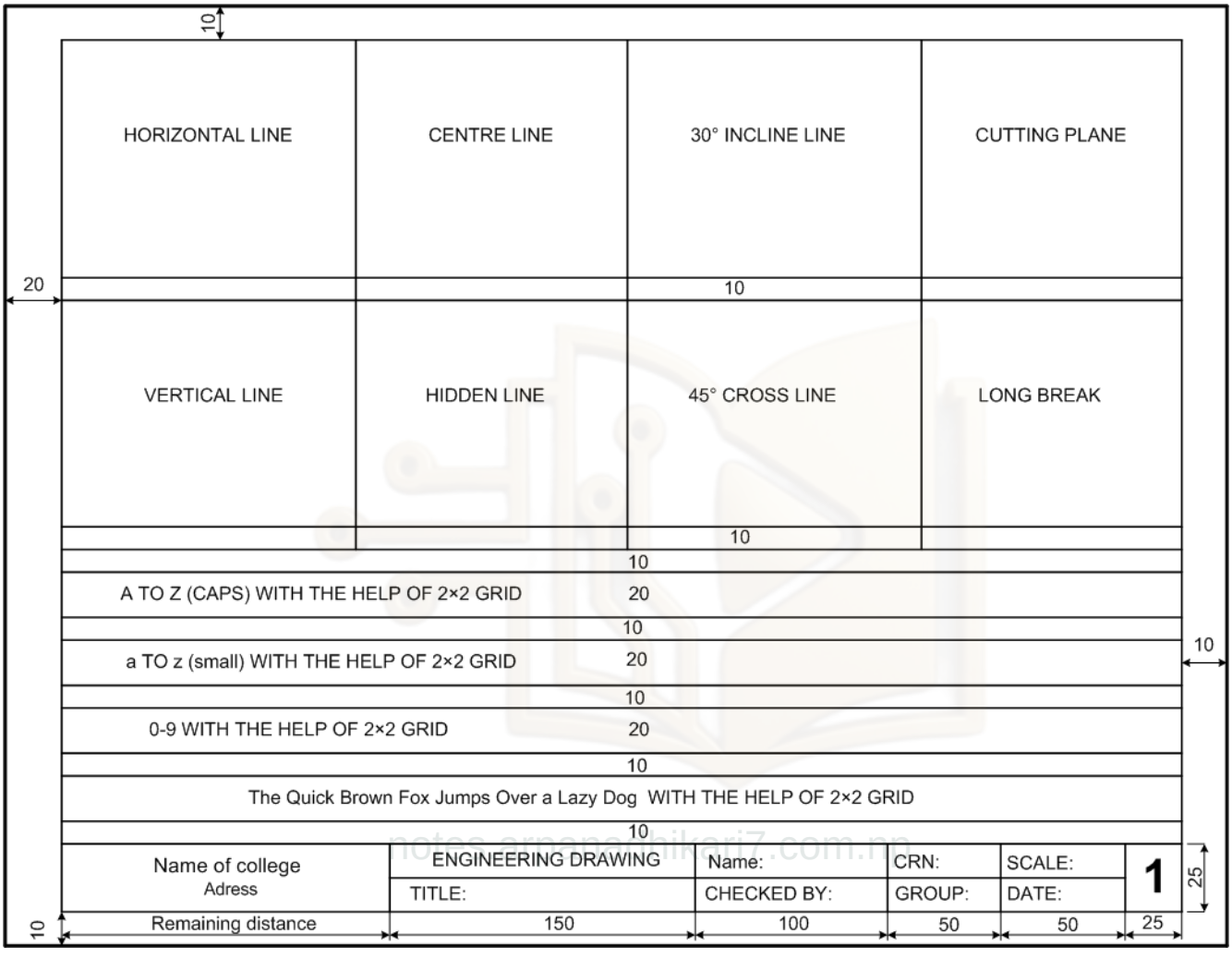
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Sheet No 1: Instrumental Drawing and Lettering

Draw the following types of line with the space between each line to be 4 mm

(a) Horizontal line (b) Vertical line (c) Centre line (d) 30° incline line
(e) 45° cross line (f)Hidden line (g) Cutting plane (h)Long Break line

Following template can be used for drawing lines and for free hand lettering. Paper used for the template is A2 sized paper.



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Sheet No 2: Dimensioning

Reproduce the following figures with appropriate dimension techniques

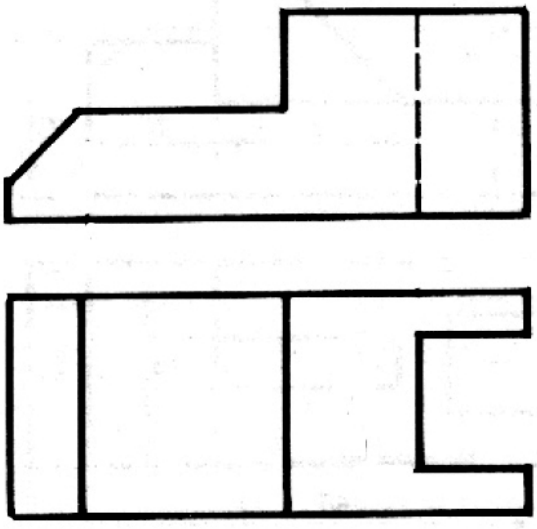


Figure 1

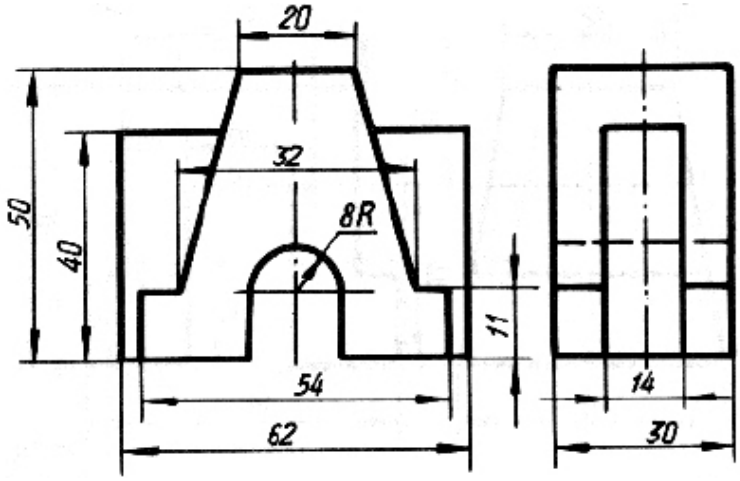


Figure 2

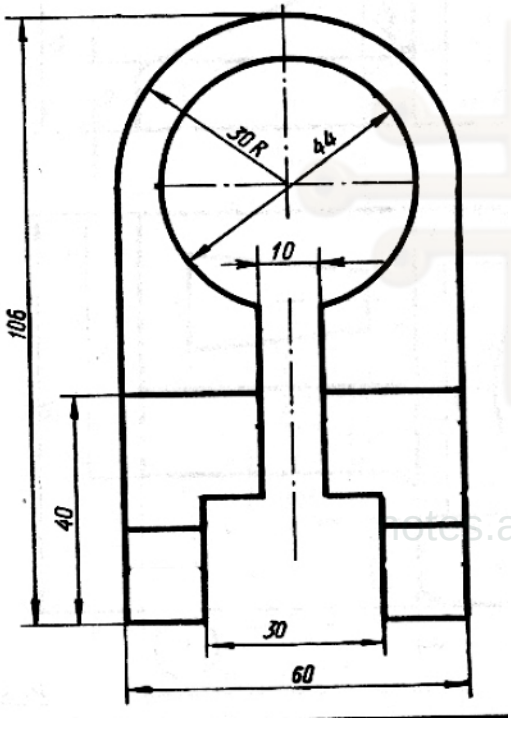


Figure 3

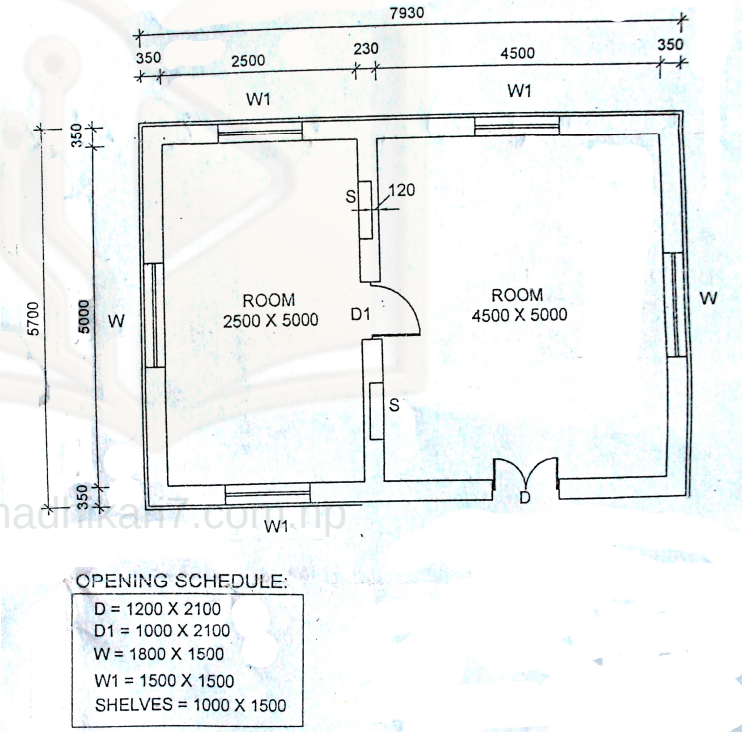


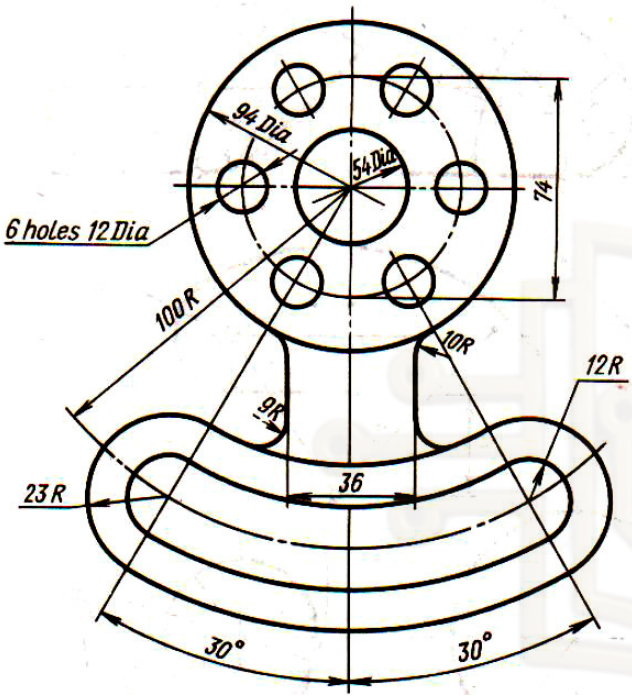
Figure 4

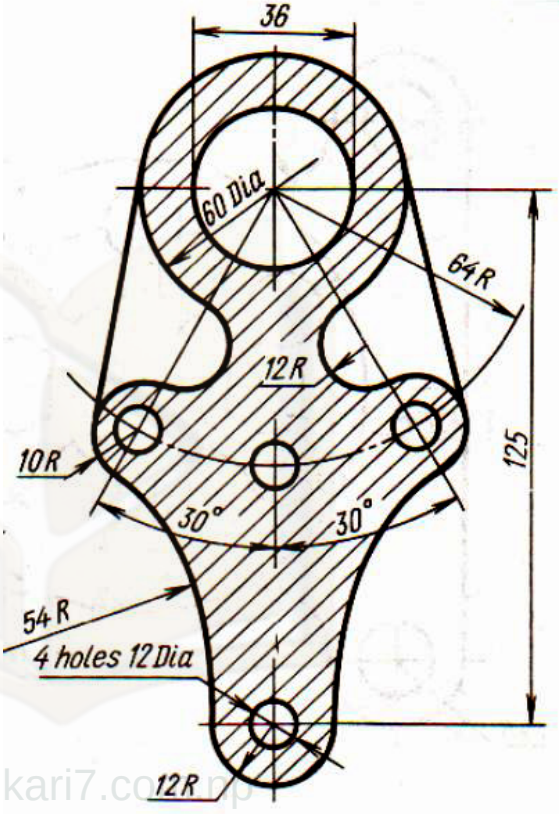
For ***1*** measure the figure and draw in the convenient scale. If any dimension is missing in figure ***3*** find scale of drawing and use it to find the missed dimension. For figure ***4*** use building dimensioning techniques.

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Sheet No 3: Applied Geometry

1. Draw a line whose length is greater than 50 mm and divide it into 7 equal parts.
2. Draw the following regular polygon of side 30 mm by *square method,*
   a. Pentagon b. Hexagon c. Heptagon
3. Draw the following regular polygon by *circle method* where the radius of circumscribing circle is 30 mm.
   a. Pentagon b. Heptagon c. Octagon
4. Draw an involute of a square having side of 15 mm.
5. Draw an involute of a circle having diameter of 20 mm.
6. Reproduce the following figures.





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Sheet No 4: Applied Geometry II

1. Draw an ellipse having eccentricity 35/45 (e<1), where its focus lies 15 mm from the directrix.
2. Draw a parabola having eccentricity 45/45 (e=1), where its focus lies 15 mm from the directrix.
3. Draw a hyperbola having eccentricity 45/35 (e>1), where its focus lies 15 mm from the directrix.
4. Draw an ellipse having major axis 90 mm and minor axis 60 mm by the concentric circle method.
5. Draw an ellipse by four centre method with radius of guide circle as 20 mm and distance between their centre as 90 mm
6. Construct one turn of conical helix with diameter if 30 mm and pitch of 60 mm.
7. Construct one turn of conical helix with base diameter 30 mm and pitch of 60 mm.
8. Draw an Archemedian's spiral of one convolution of radius 100 mm.

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Sheet No 5: Orthographic Drawing I

Draw the three-view orthographic drawing projection of the following objects.

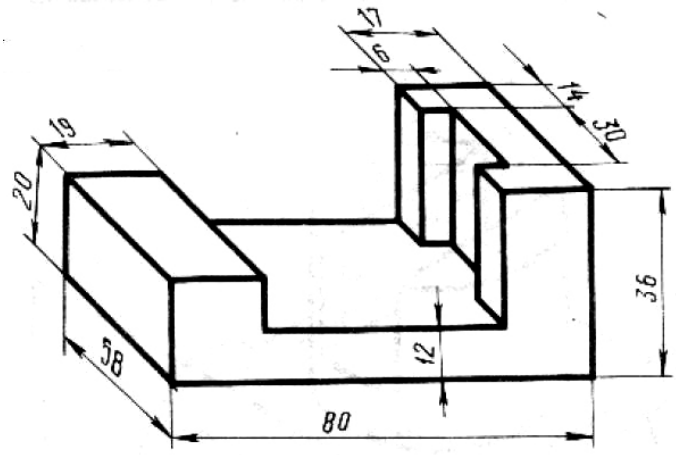


Figure 1

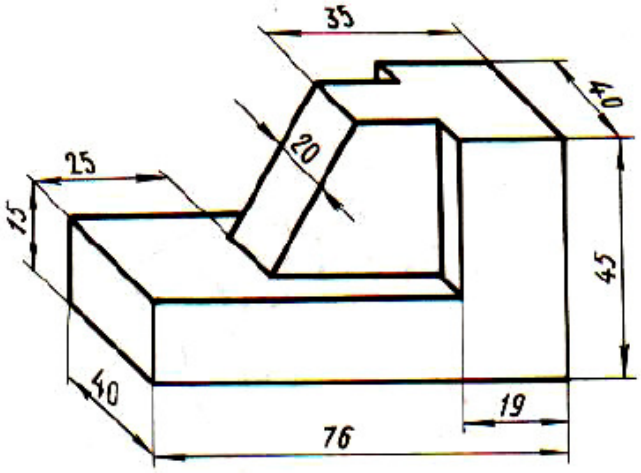


Figure 2

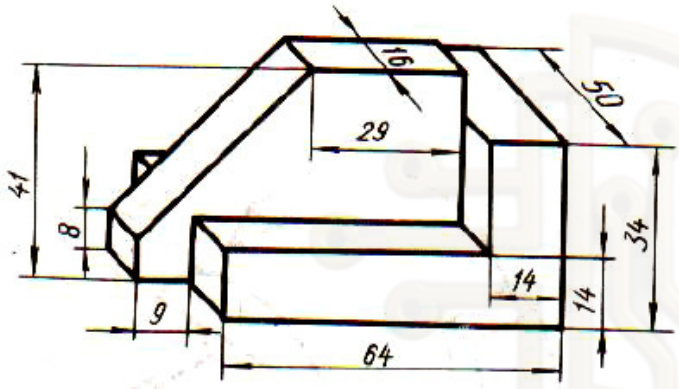


Figure 3

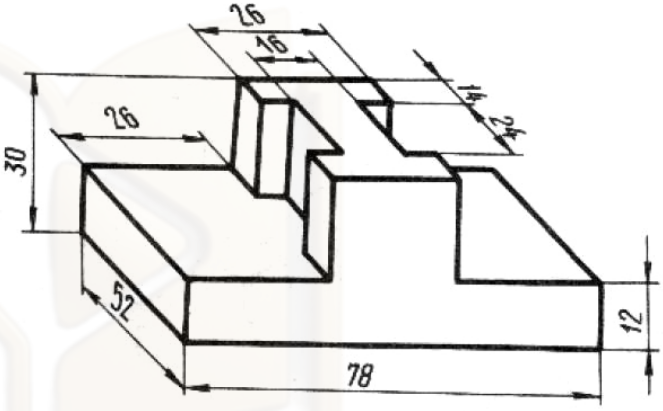


Figure 4

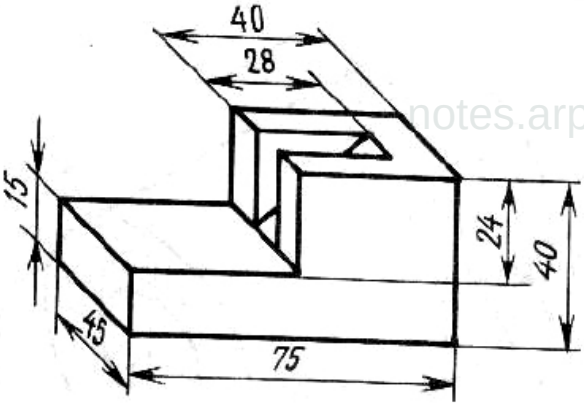


Figure 5

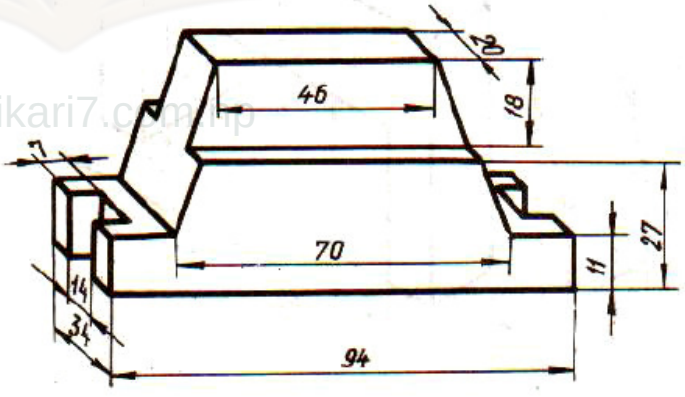


Figure 6

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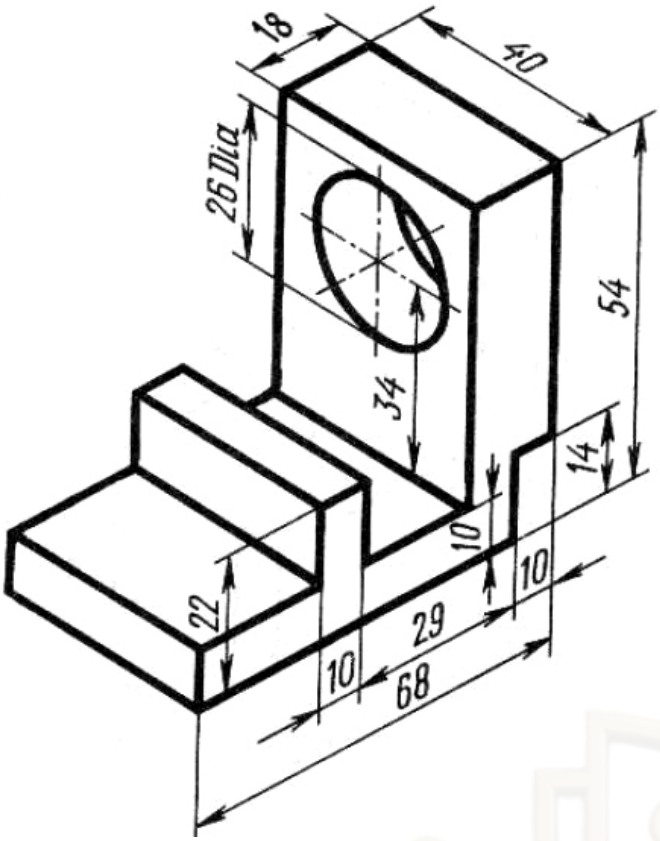


Figure 7

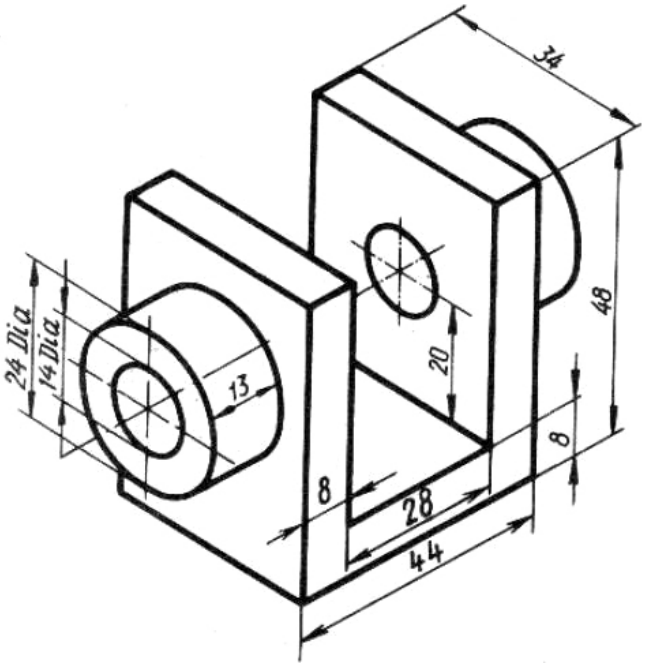


Figure 8

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Sheet No 6: Orthographic Drawing II

Draw the three-view orthographic drawing projection of the following objects.

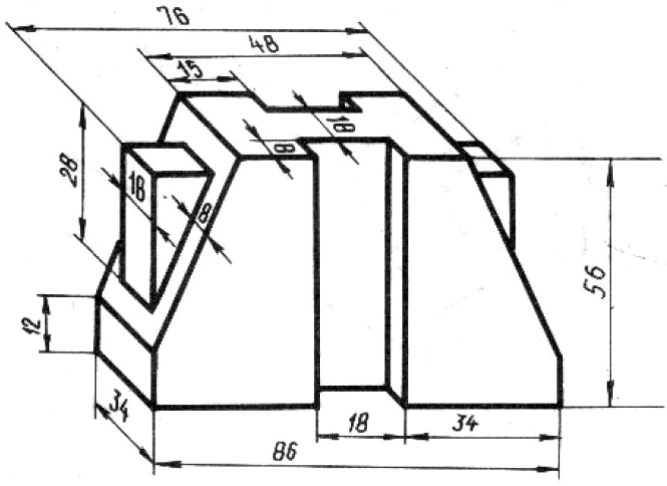


Figure 1

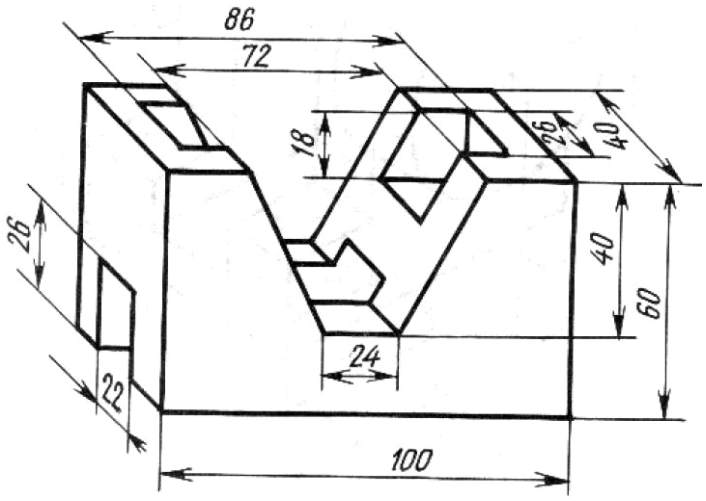


Figure 2

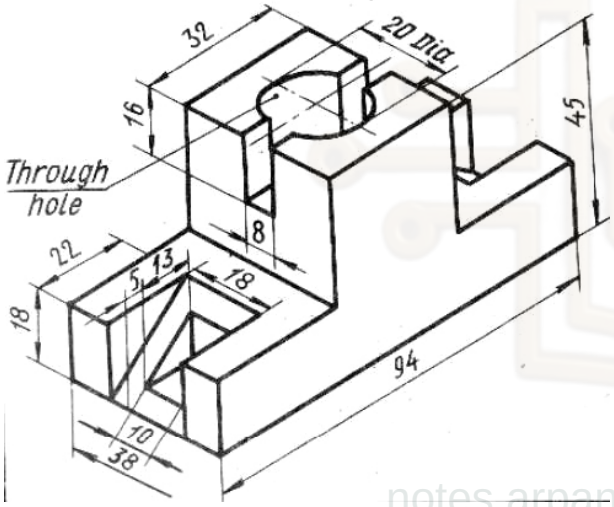


Figure 3

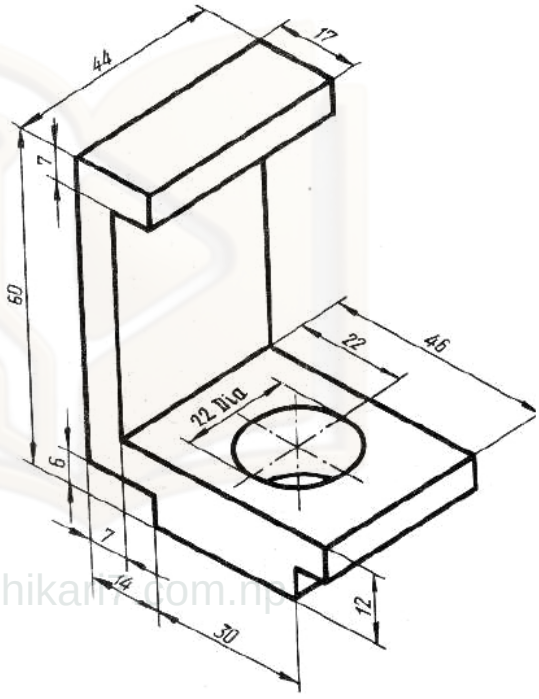


Figure 4

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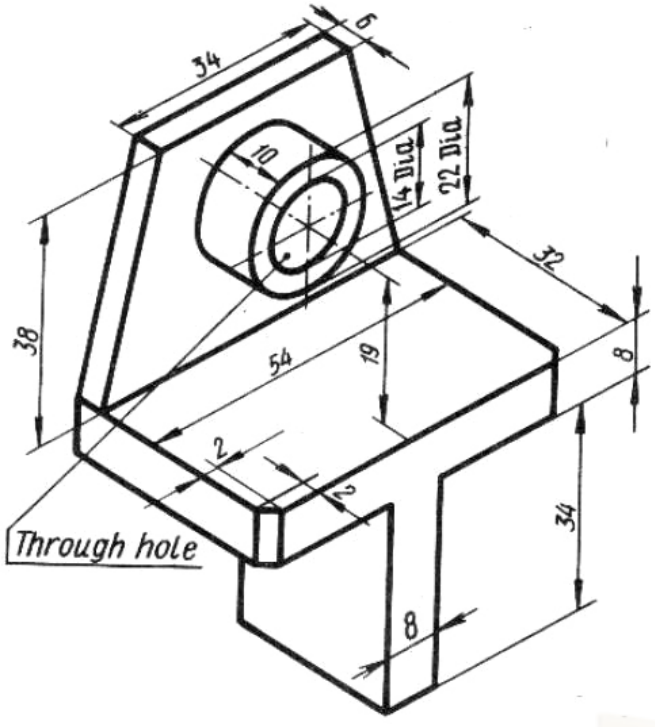


Figure 5

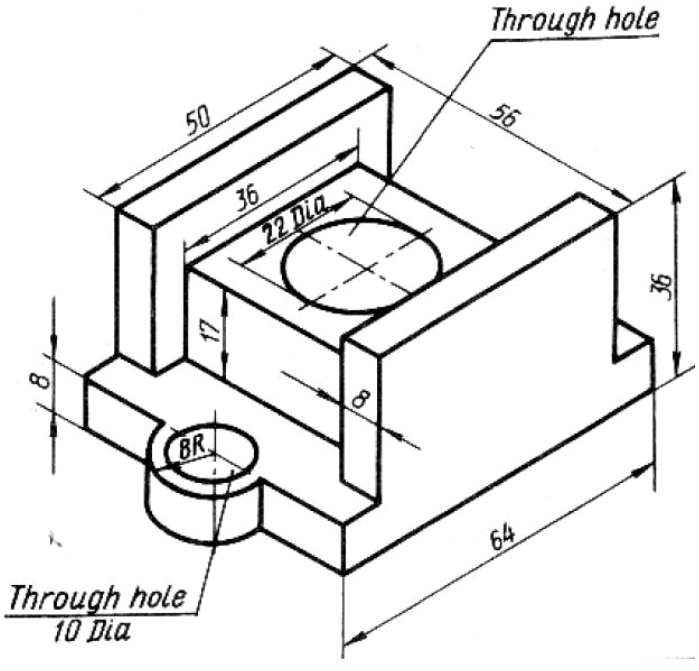


Figure 6

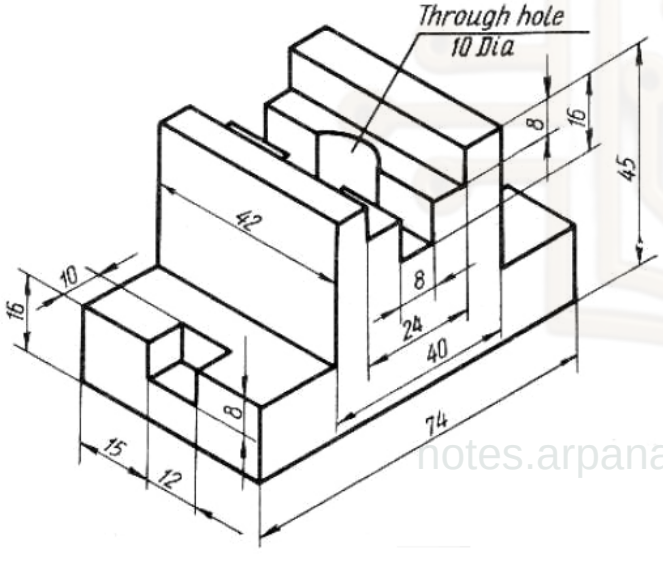


Figure 7

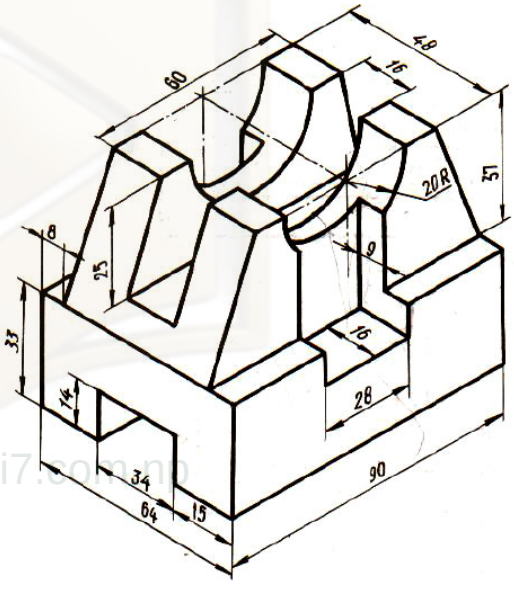


Figure 8

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Sheet No 7: Sectional view and auxiliary view

Make the three-view orthographic drawing with required sectional view of the following objects

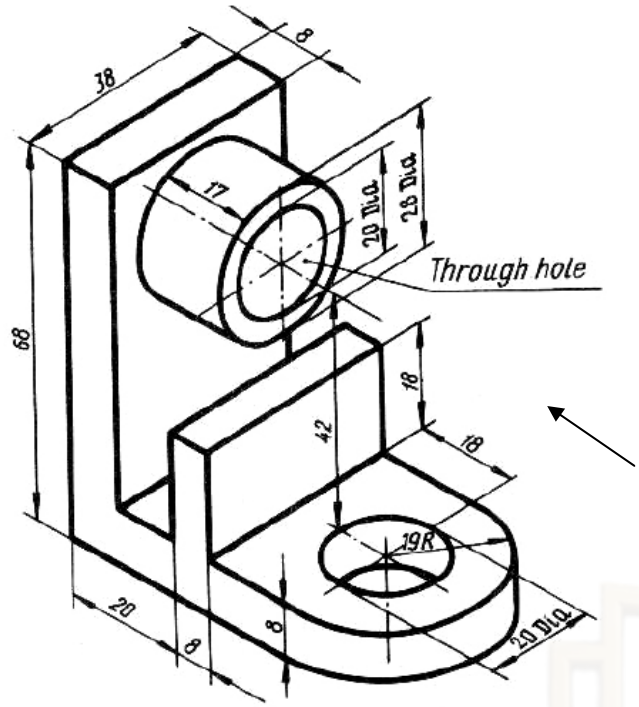


Figure 1 (Sectional side view)

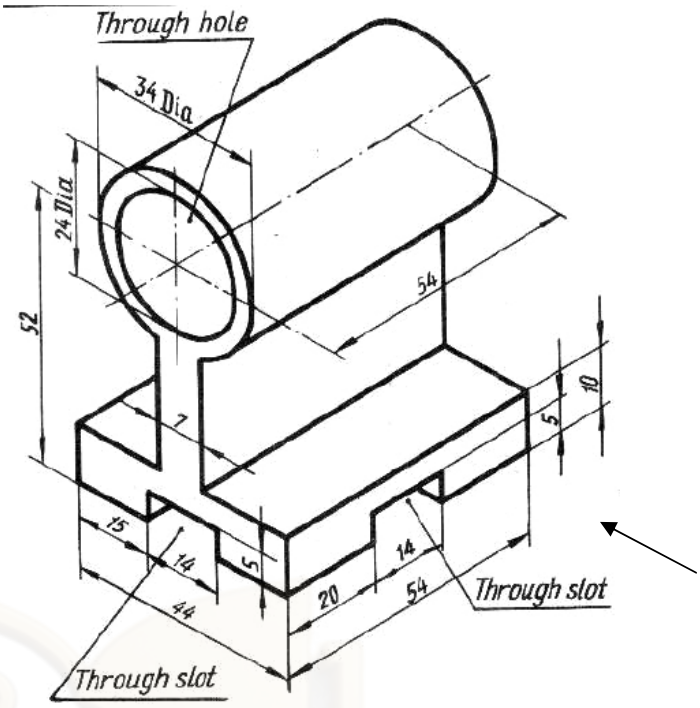


Figure 2 (Sectional front view)

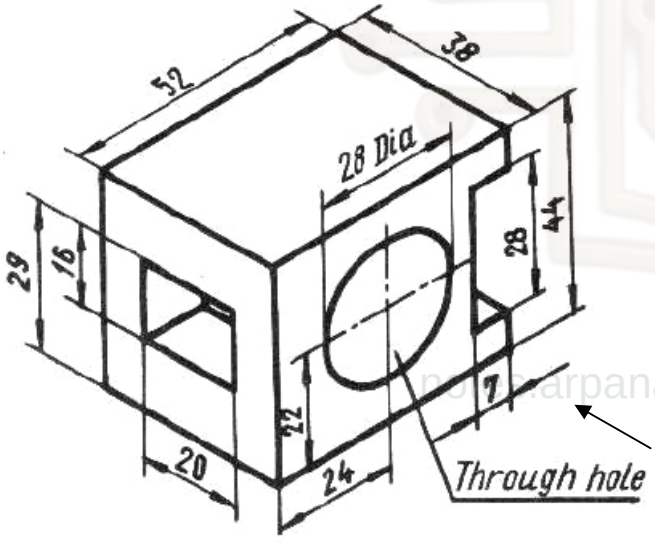


Figure 3 (Sectional side view)

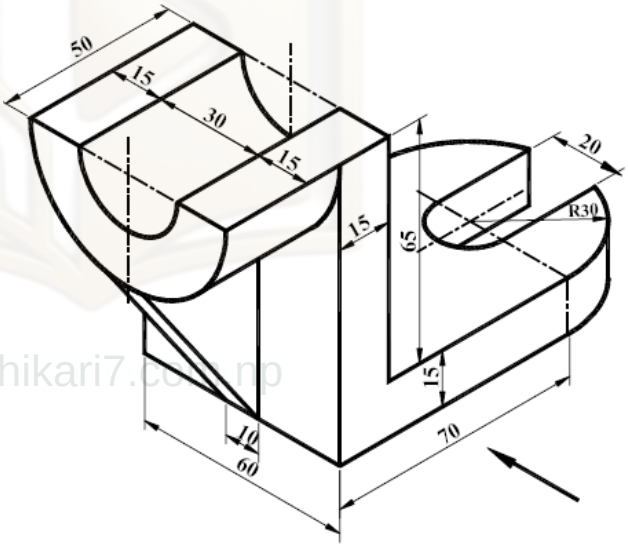


Figure 4 (Sectional front view)

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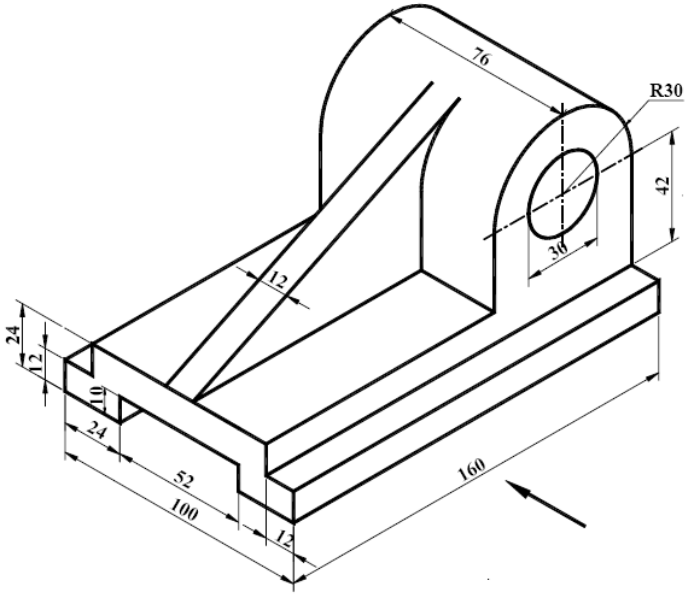


Figure 5 (Sectional front view)

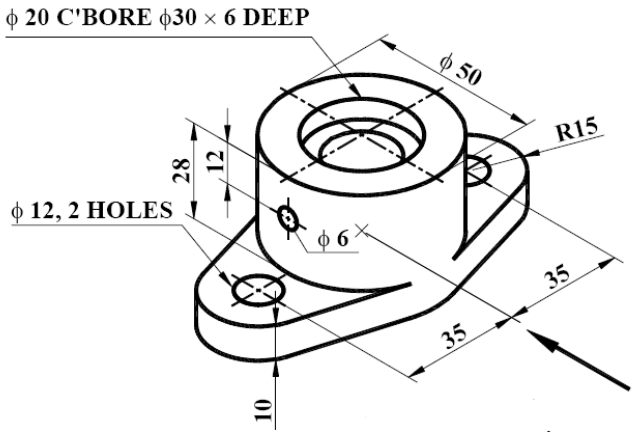


Figure 6 (half sectional front view, half sectional side view and top view)

Make complete orthographic views with auxiliary view for the inclined surface

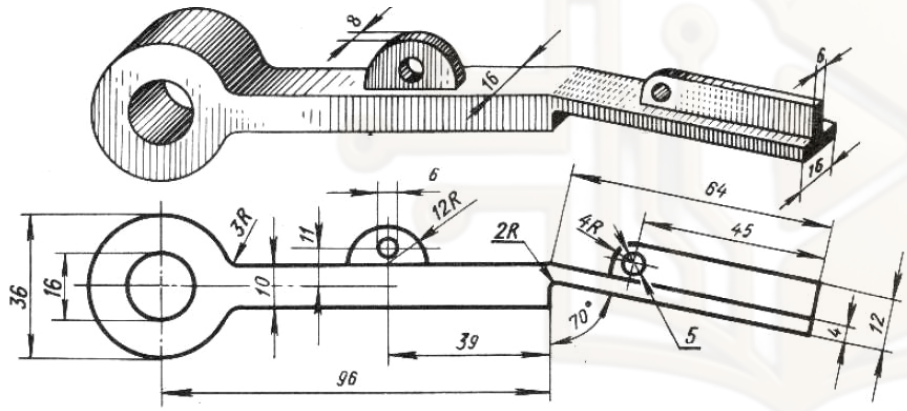


Figure 7

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Sheet No 8: Isometric Drawing and Oblique Drawing

Draw the isomeric drawings from the following orthographic drawings

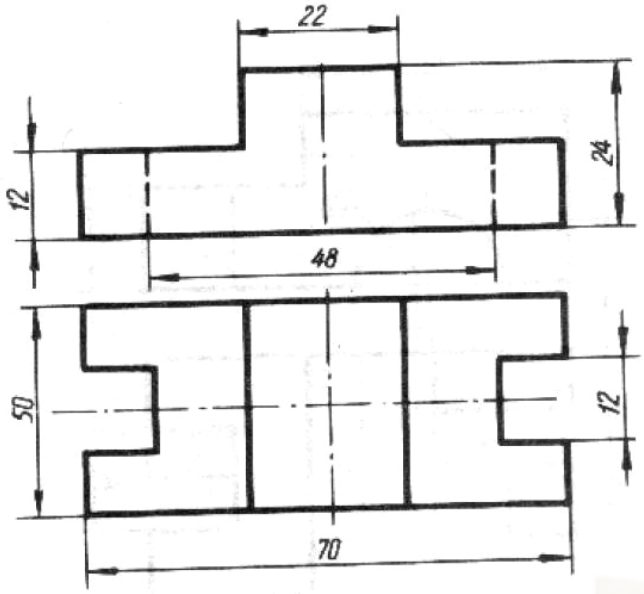


Figure 1

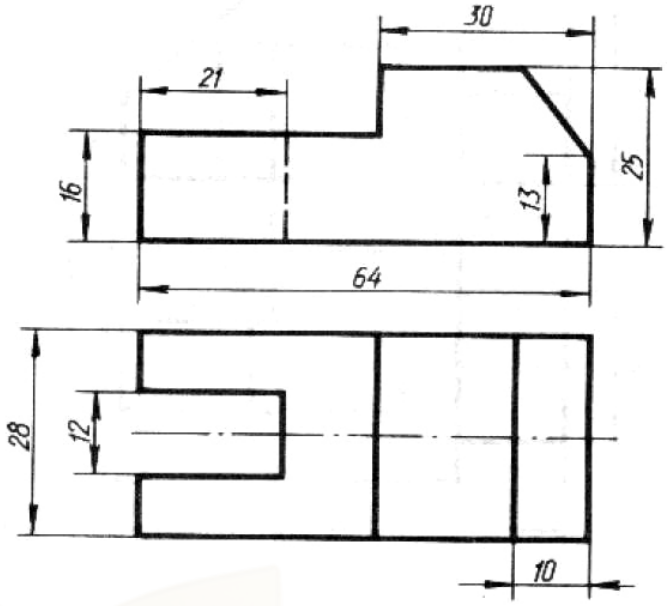


Figure 2

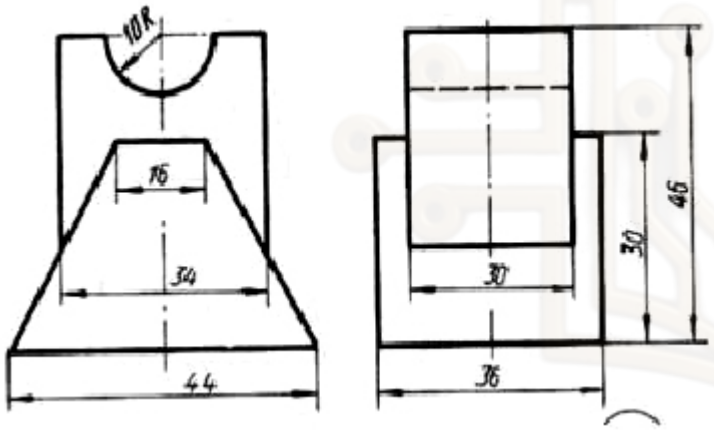


Figure 3

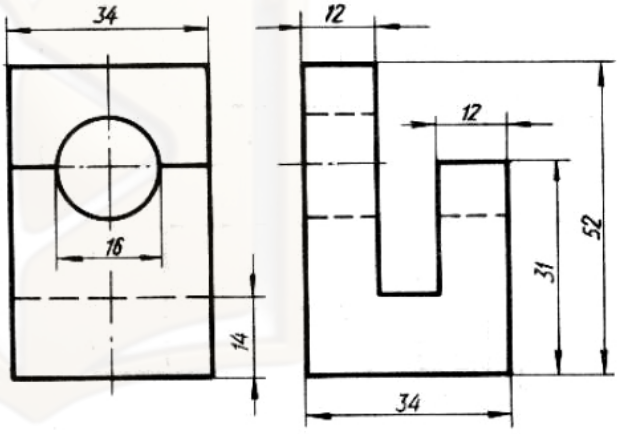


Figure 4

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Draw the oblique drawings from the following orthographic drawings

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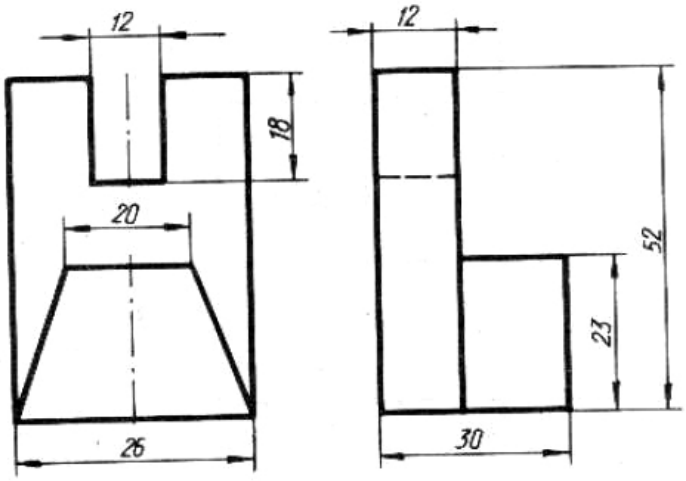


Figure 5

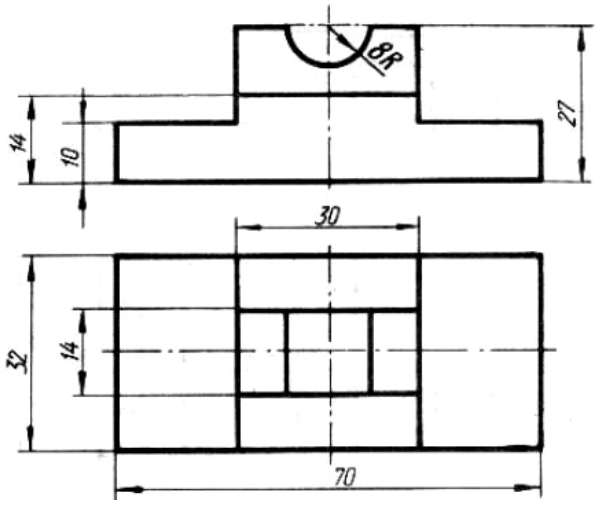


Figure 6

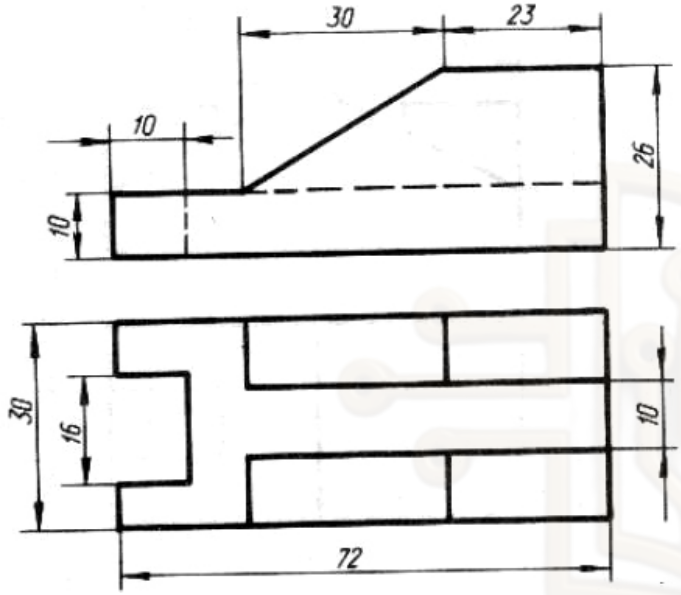


Figure 7

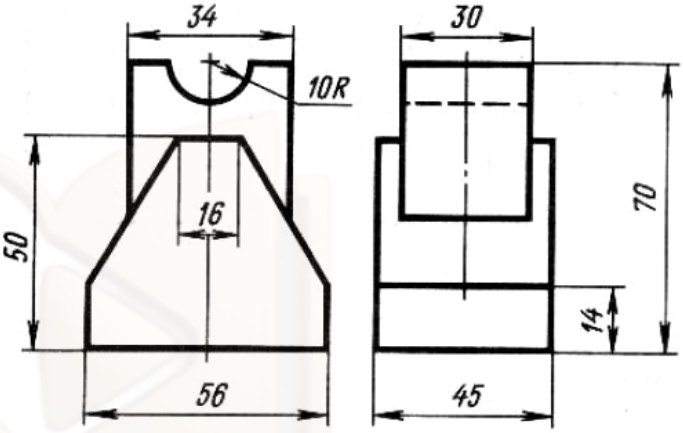


Figure 8

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Sheet No 9: Descriptive Geometry

1. Draw the pictorial and orthographic projection of a line CD, 15 mm long, perpendicular to HP and 20 mm away from VP when one of its extremities near to HP is 10 mm away from the HP.
2. A Square lamina ABCD of 15 mm side is parallel to HP and is 10 mm in front of it. Its side near to VP is parallel to and 10 mm from VP. Draw its pictorial and orthographic projections.
3. A square lamina ABCD of 30 mm side is perpendicular to VP and inclined to HP at 45°. It rests on its side BC in HP. Draw its projections when corner point C is 12 mm in front of the VP.
4. Reproduce the given views of the lines and determine the true lengths using the auxiliary view method and revolution method.

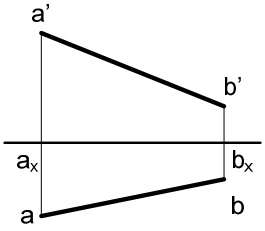


Figure 4a

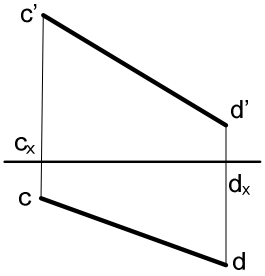


Figure 4b

$a'a_x = 26$	$b'b_x = 10$
$aa_x = 22$	$bb_x = 15$
$a_xb_x = 48$	

$c'c_x = 27$	$d'd_x = 7$
$cc_x = 5$	$dd_x = 25$
$c_xd_x = 48$	

5. Determine the shortest distance between the line AB and the point C

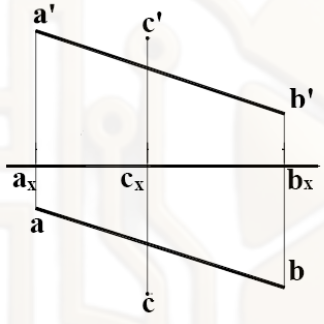


Figure 5

$a'a_x = 28$	$a'a = 5$
$b'b_x = 8$	$bb_x = 25$
$c'c_x = 26$	$cc_x = 26$
$a_xc_x = 23$	$c_xb_x = 26$

6. Reproduce the given views of the plane and determine the true shape of the plane

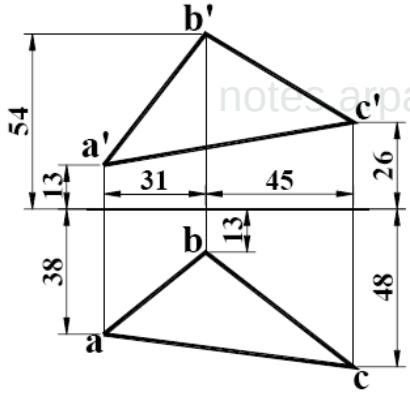


Figure 6

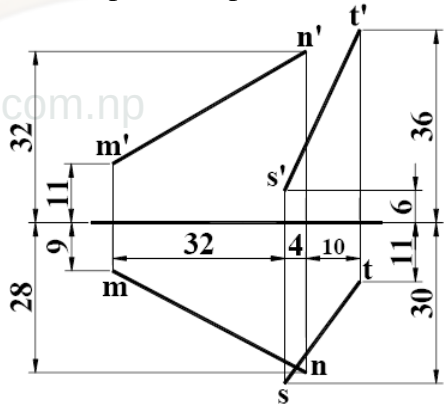


Figure 7

7. Determine the shortest distance between two lines MN and ST

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Sheet No 10: Development of surfaces

Make a complete orthographic drawing of a geometrical solid cut by a plane **P**. Find the true shape of the section. Construct the development of the surface of the solid.

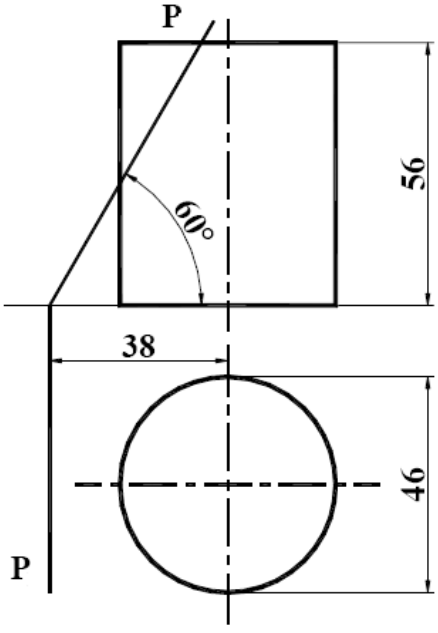


Figure 1

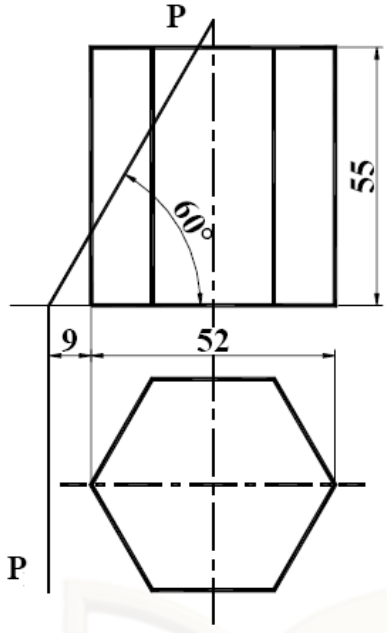


Figure 2

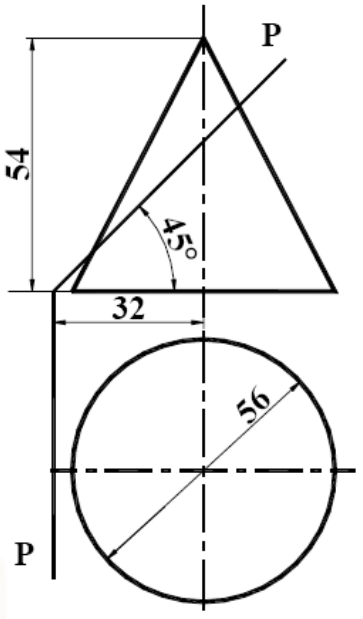


Figure 3

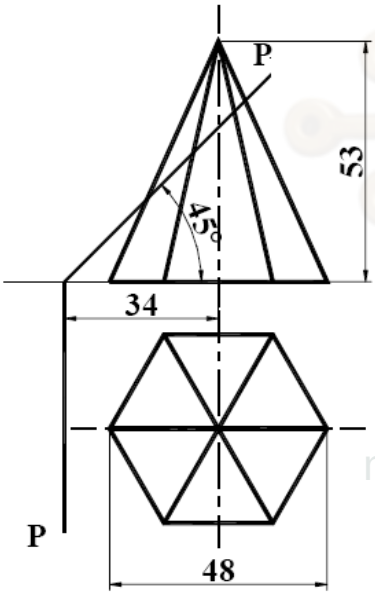


Figure 4

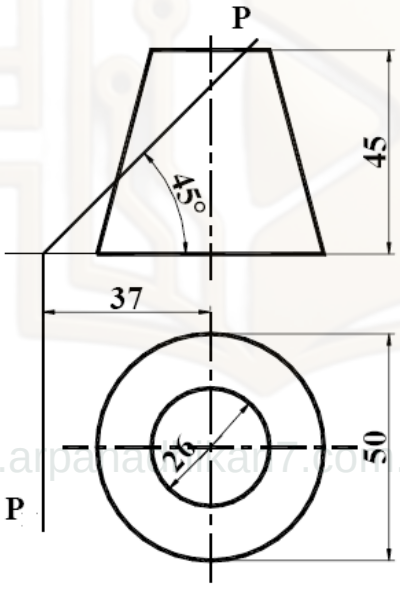


Figure 5

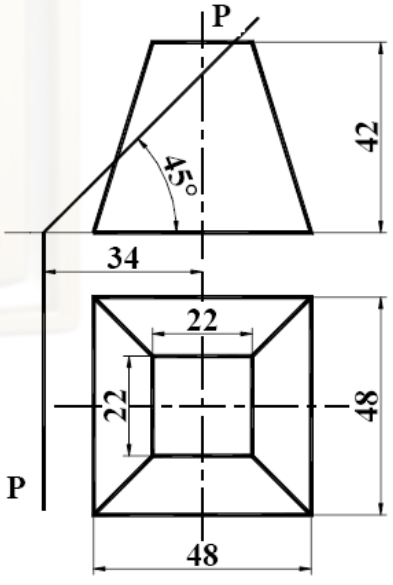


Figure 6

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Sheet No 11: Intersection of solids

Draw the given views of and with the curve of intersection. Also develop the lateral surface of each solid.

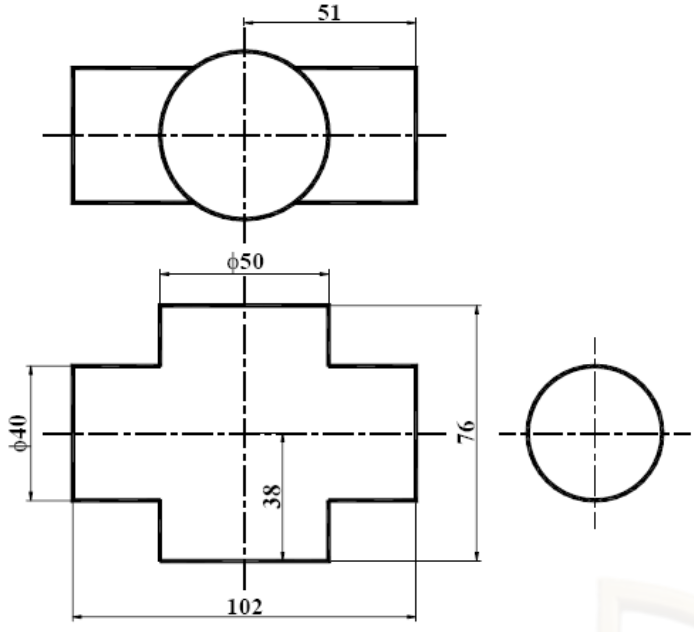


Figure 1

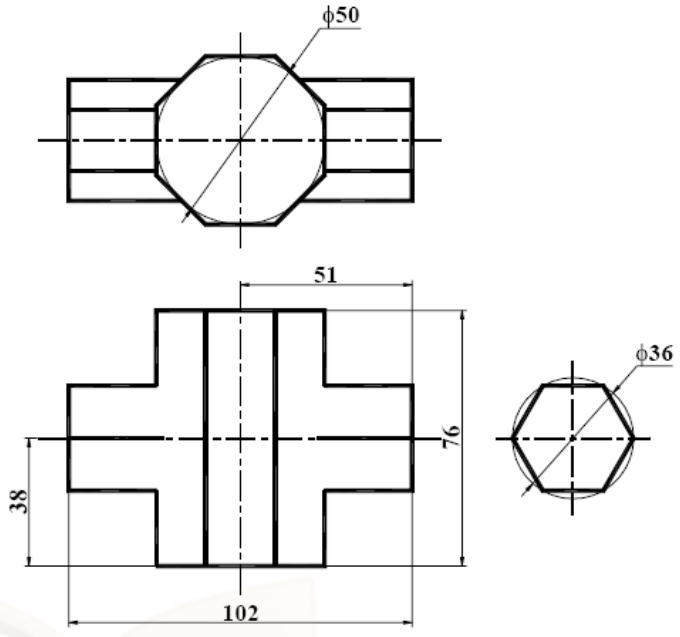


Figure 2

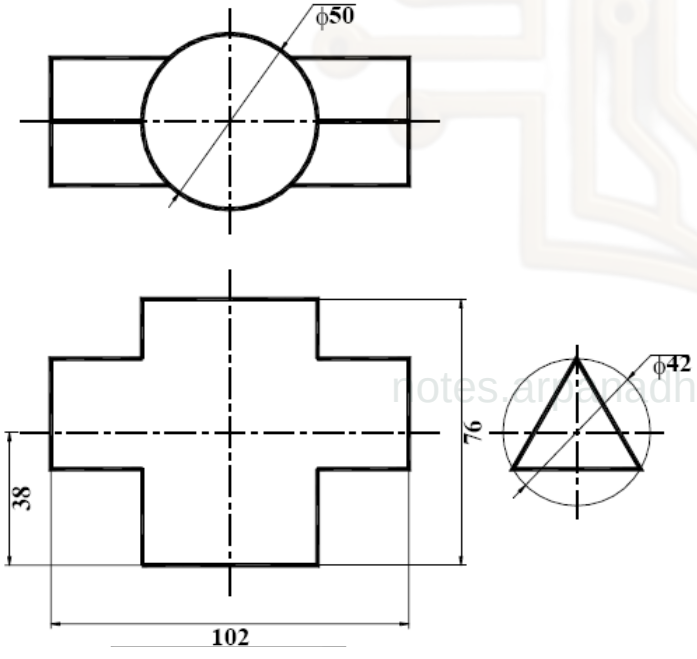


Figure 3

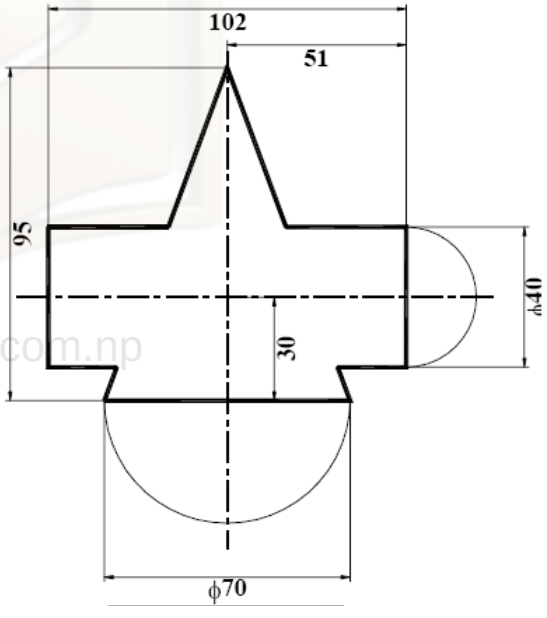


Figure 4

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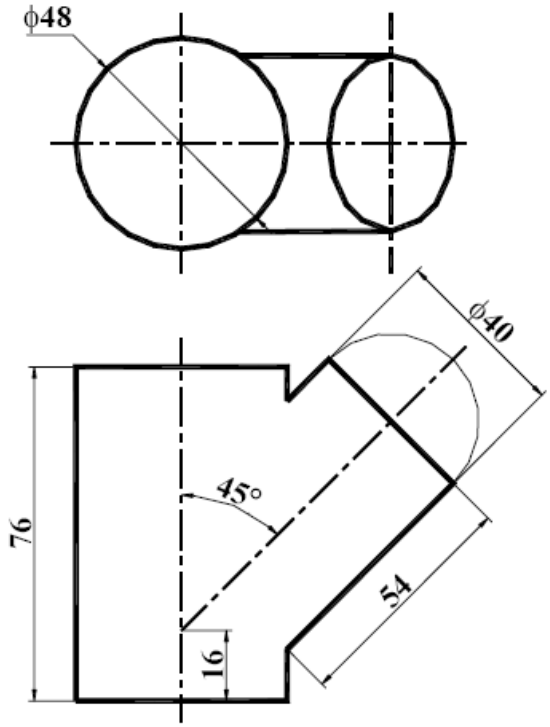


Figure 5

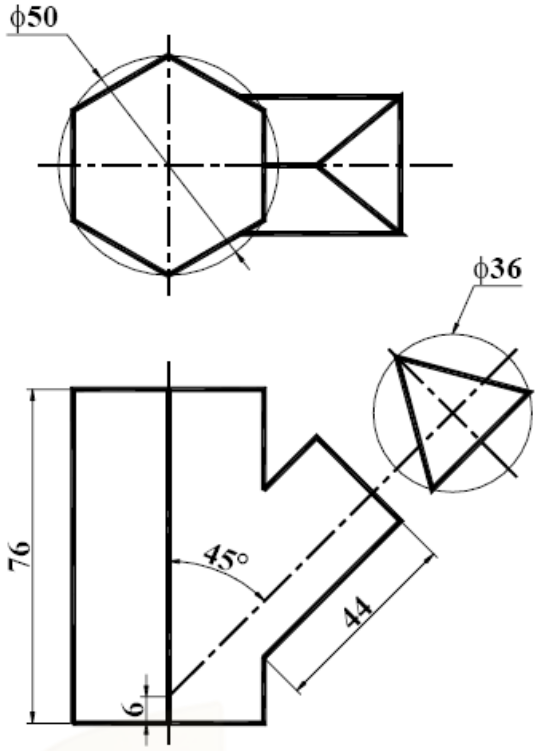


Figure 6

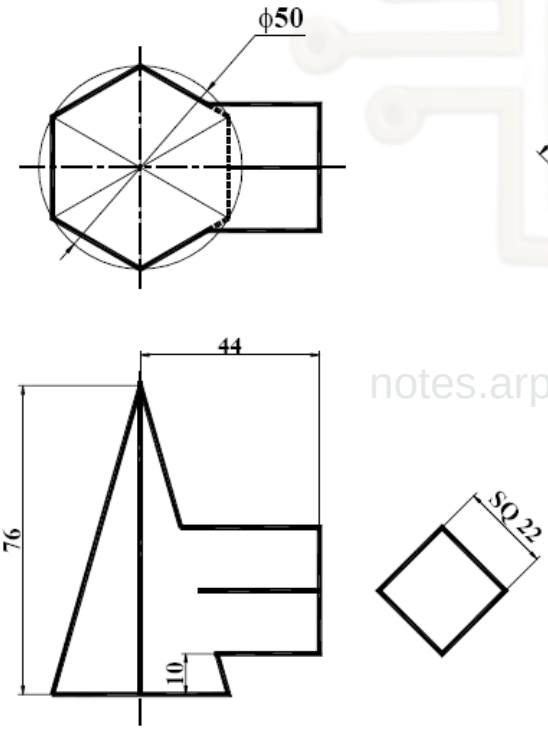


Figure 7

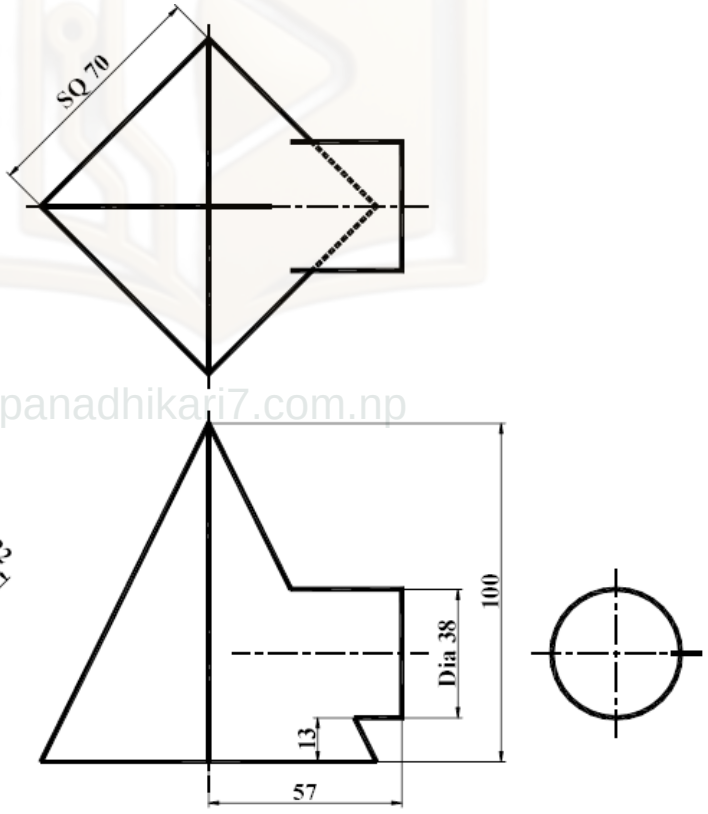


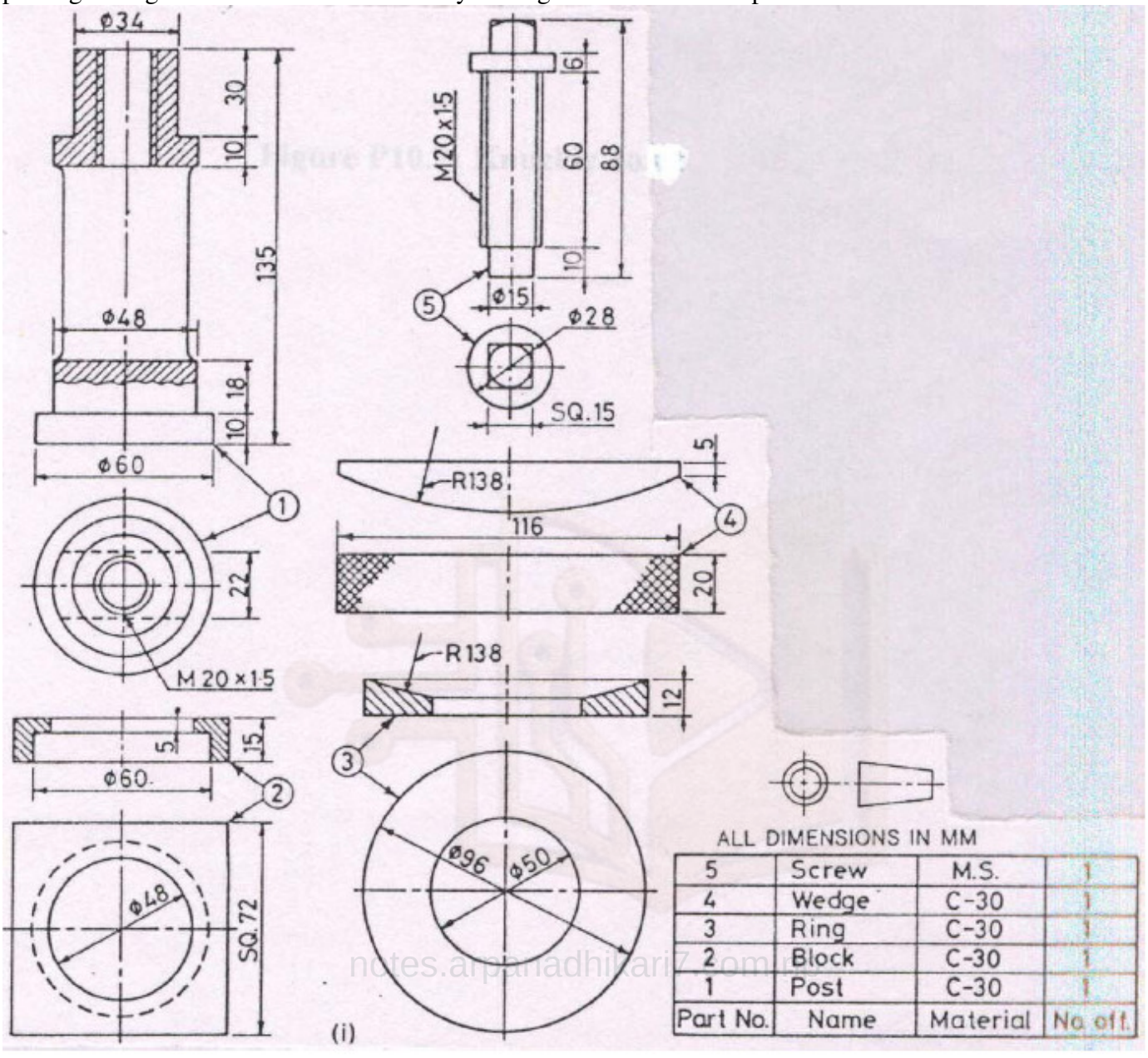
Figure 8

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Sheet No 12: Assembly drawing I

Assemble all parts of the tool post given below and draw its sectional front view, with the section plane passing through the vertical slot in the body and right side view and top view.

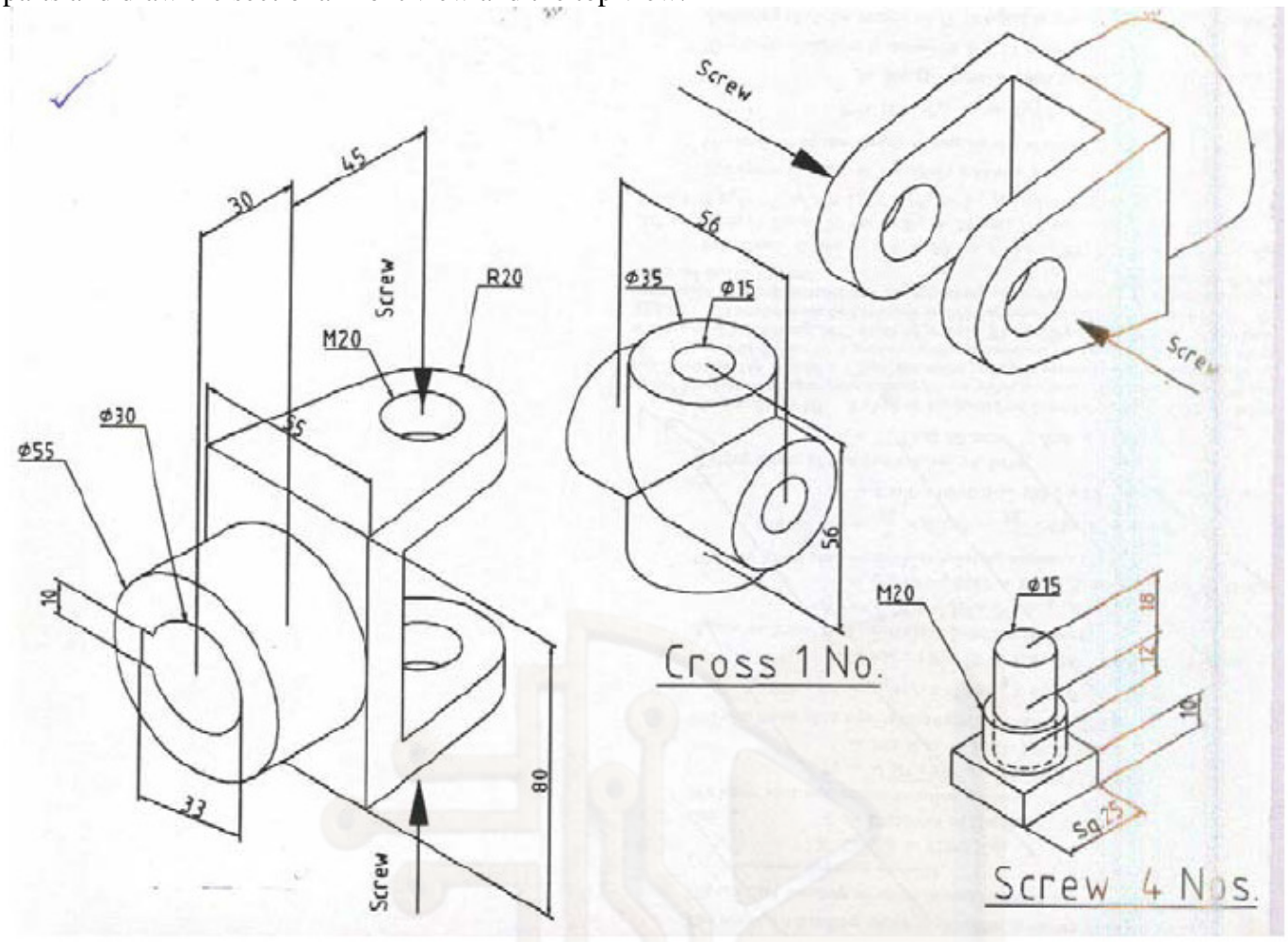


Part No.	Name	Material	No. off.
5	Screw	M.S.	1
4	Wedge	C-30	1
3	Ring	C-30	1
2	Block	C-30	1
1	Post	C-30	1

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Figure below shows the detail drawing of a Universal Coupling for connecting two shafts. Assemble the parts and draw the sectional front view and the top view.

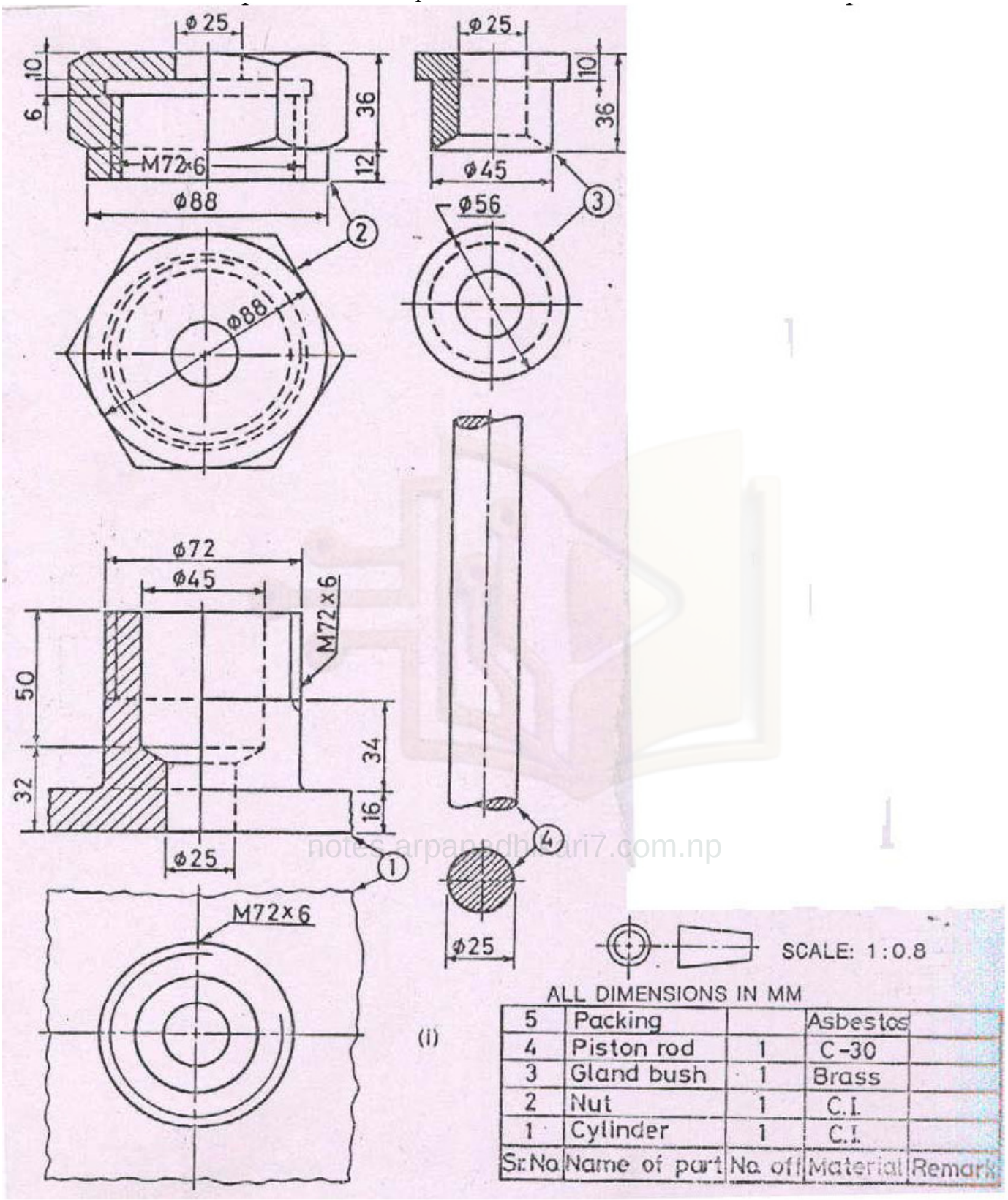


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Sheet No 13: Assembly drawing II

Figure below shows the detail drawing of a Stuffing Box for a small steam engine. Assemble the parts and draw the half sectional front view and the top view.



Sr.No	Name of part	No. of	Material	Remark
5	Packing		Asbestos	
4	Piston rod	1	C-30	
3	Gland bush	1	Brass	
2	Nut	1	C.I.	
1	Cylinder	1	C.I.	

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